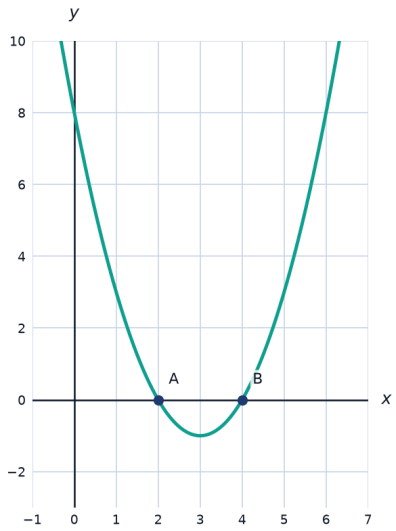
Modulus Functions & the Factor/Remainder Theorem

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

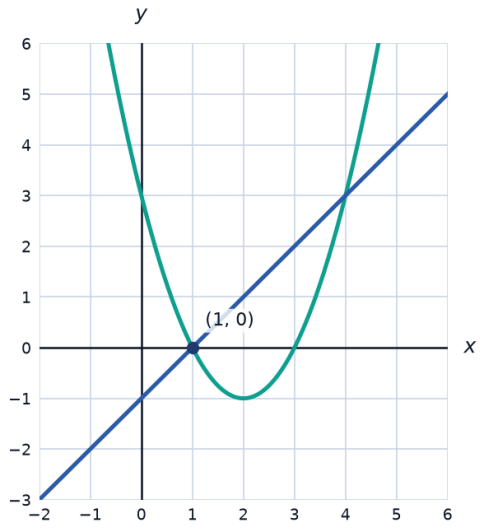
1 **[Foundational]** • **Target: Solving Modulus Equations**

Consider the function $f(x) = x^2 - 6x + 8$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at points A and B. What are the coordinates of point A? Show your complete algebraic work.



2 **[Foundational]** • **Target: Modulus Inequalities**

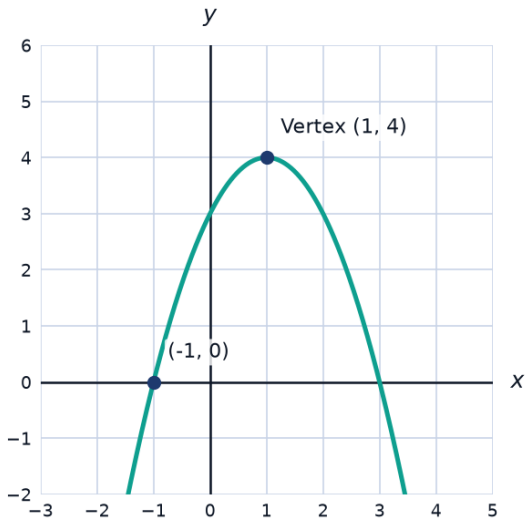
The diagram shows the graph of the function $y = f(x)$. The function is defined by $f(x) = x^2 + bx + c$. The graph passes through the point $(1, 0)$ and has a vertex at $(2, -1)$. Find the value of b.



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3 [Foundational] • Target: Polynomial Remainder & Factor Theorems

The diagram shows the graph of the quadratic function $f(x) = ax^2 + bx + c$. The vertex of the parabola is at $(1, 4)$, and it passes through the point $(-1, 0)$. Find the value of b.



4 [Foundational] • Target: Transformations of Modulus Graphs

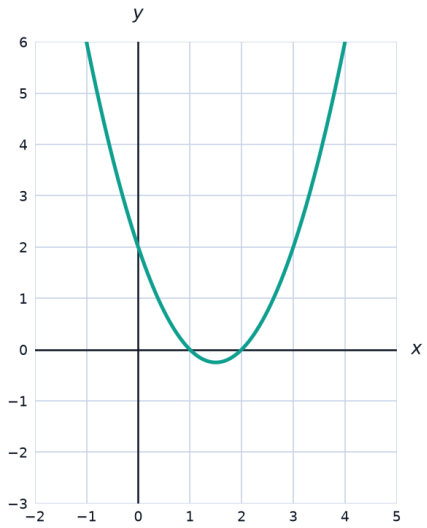
The diagram shows the graph of the quadratic function $f(x) = ax^2 + bx - 2$. The graph passes through the points $(1, -3)$ and $(2, 2)$. Determine the value of a.



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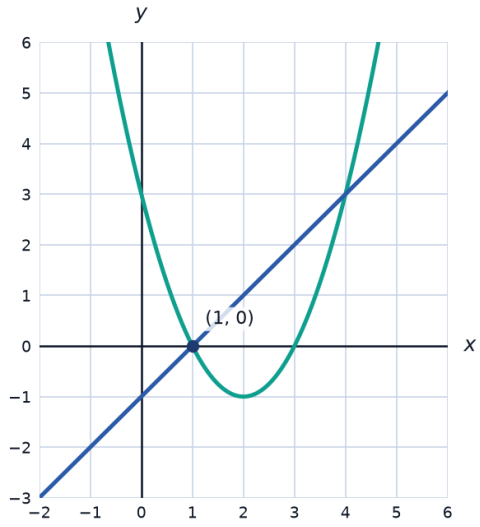
5 [Foundational] • Target: Solving Modulus Equations

Consider the function $f(x) = x^2 - 3x + 2$. The graph of $y = f(x)$ is a parabola. Find the x-coordinates of the points where the graph intersects the x-axis.



6 [Foundational] • Target: Modulus Inequalities

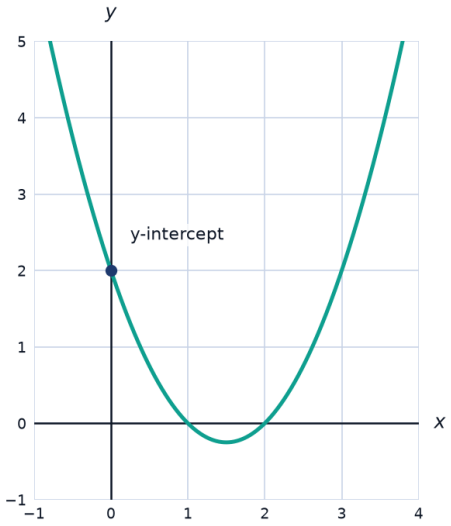
The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points A and B. The vertex of the parabola is at point C. Determine the coordinates of the point where the graph intersects the y-axis.



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7 [Foundational] • Target: Polynomial Remainder & Factor Theorems

Consider the function $f(x) = x^2 - 3x + 2$. The graph of $y = f(x)$ is shown. Determine the y-coordinate of the y-intercept.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Transformations of Modulus Graphs

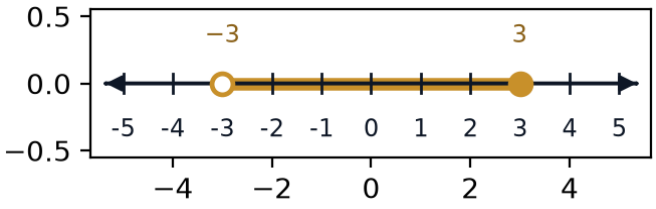
Consider the function $f(x) = \frac{ax^2 + bx + c}{dx + e}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 2$ and a horizontal asymptote at $y = 3$. The graph also passes through the points $(0, 0)$ and $(1, -3)$. Find the value of $\frac{a}{d}$.

9 [Intermediate] • Target: Solving Modulus Equations

Consider the function $f(x) = \frac{ax^2 + bx + c}{x - d}$, where $a, b, c, d \in \mathbb{R}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 3$ and a horizontal asymptote at $y = 2$. The function passes through the point $(1, -8)$. If $f'(x)$ has a critical point at $x = 5$, find the value of d.

10 [Intermediate] • Target: Modulus Inequalities

Determine the equation of the oblique asymptote to the graph of $f(x) = \frac{2x^2 - 3x + 4}{x - 1}$.



11 [Intermediate] • Target: Polynomial Remainder & Factor Theorems

Evaluate $\cos\left(\sin^{-1}\left(-\frac{1}{2}\right)\right)$.

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12 [Intermediate] • Target: Transformations of Modulus Graphs

Evaluate $\cos\left(\sin^{-1}\left(\frac{1}{2}\right)\right)$.

13 [Intermediate] • Target: Solving Modulus Equations

Find the oblique asymptote of $f(x) = \frac{4x^2 + 3x - 14}{(x + 2)}$, in the form $y = mx + c$.

14 [Intermediate] • Target: Modulus Inequalities

Find the oblique asymptote of $f(x) = \frac{3x^2 - 10x + 11}{(x - 2)}$, in the form $y = mx + c$.

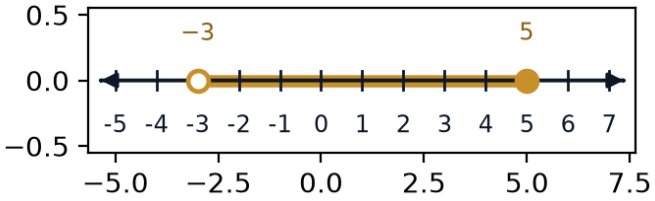
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Polynomial Remainder & Factor Theorems • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Find the oblique asymptote of $f(x) = \frac{2x^2 + 4x - 20}{(x + 4)}$, in the form $y = mx + c$.

16 [Challenging] • Target: Transformations of Modulus Graphs • Cross-Lesson: 1.04 Partial Fractions

Find the oblique asymptote of $f(x) = \frac{4x^2 - 11x + 2}{(x - 2)}$, in the form $y = mx + c$.



17 [Challenging] • Target: Solving Modulus Equations • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Find the oblique asymptote of $f(x) = \frac{4x^2 - 14x + 7}{(x - 3)}$, in the form $y = mx + c$.

18 [Challenging] • Target: Modulus Inequalities • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Find the oblique asymptote of $f(x) = \frac{4x^2 + 19x - 2}{(x + 5)}$, in the form $y = mx + c$.

19 [Challenging] • Target: Polynomial Remainder & Factor Theorems • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Find the oblique asymptote of $f(x) = \frac{2x^2 + 9x + 8}{(x + 4)}$, in the form $y = mx + c$.

20 [Challenging] • Target: Transformations of Modulus Graphs • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Find the oblique asymptote of $f(x) = \frac{2x^2 - 4x + 3}{(x - 1)}$, in the form $y = mx + c$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Solving Modulus Equations

Solve the equation $|2x - 5| = x + 1$.

a. $x = 6$ and $x = \frac{4}{3}$

b. $x = 6$ only

c. $x = \frac{4}{3}$ only

d. $x = -6$ and $x = -\frac{4}{3}$

22 [Foundational] • Target: Solving Modulus Equations

Solve $|3x + 2| = |x - 4|$.

a. $x = \frac{1}{2}$ and $x = -3$

b. $x = 2$ and $x = -3$

c. $x = -\frac{1}{2}$ and $x = 3$

d. $x = 1$ and $x = -4$

23 [Foundational] • Target: Solving Modulus Equations

Why can $|x - 7| = -3$ have no real solutions?

a. The absolute value of any real expression is non-negative ($|u| \geq 0$), so it cannot equal a negative number

b. Because x must be greater than 7

c. Because -3 is not a prime number

d. The equation has complex solutions only

24 [Foundational] • Target: Modulus Inequalities

Solve the inequality $|2x - 3| < 7$.

a. $-2 < x < 5$

b. $x < 5$

c. $x < -2$ or $x > 5$

d. $-5 < x < 5$

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25 [Foundational] • Target: Modulus Inequalities

Solve $|3x + 1| \geq 10$.

a. $x \leq -\frac{11}{3}$ or $x \geq 3$

b. $-\frac{11}{3} \leq x \leq 3$

c. $x \geq 3$

d. $x \leq -3$ or $x \geq \frac{11}{3}$

26 [Foundational] • Target: Modulus Inequalities

Which interval notation represents the solution to $|x - a| \leq \varepsilon$ where $\varepsilon > 0$?

a. $[a - \varepsilon, a + \varepsilon]$

b. $(a - \varepsilon, a + \varepsilon)$

c. $[-\varepsilon, \varepsilon]$

d. $[a, a + \varepsilon]$

27 [Foundational] • Target: Polynomial Remainder & Factor Theorems

When $P(x) = 2x^3 - 3x^2 + ax - 5$ is divided by $(x - 2)$, the remainder is 11. Find a.

a. 6

b. 4

c. -6

d. 8

28 [Foundational] • Target: Polynomial Remainder & Factor Theorems

If $(x + 3)$ is a factor of $Q(x) = x^3 + kx^2 - 7x + 6$, find k.

a. 2

b. -2

c. 3

d. -1

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Polynomial Remainder & Factor Theorems

A polynomial $f(x)$ leaves remainder 3 when divided by $(x-1)$ and remainder 7 when divided by $(x-3)$. What is the linear remainder when $f(x)$ is divided by $(x-1)(x-3)$?

a. $2x+1$

b. $2x-1$

c. $x+2$

d. $4x-1$

30 [Intermediate] • Target: Transformations of Modulus Graphs

How is the graph of $y=|f(x)|$ obtained from the graph of $y=f(x)$?

a. Any part of the graph lying below the x-axis is reflected vertically across the x-axis into the upper half-plane

b. The left half of the graph is discarded and replaced by reflection of the right half

c. The graph is shifted 1 unit upward

d. The graph is reflected across the y-axis

31 [Intermediate] • Target: Transformations of Modulus Graphs

How is the graph of $y=f(|x|)$ obtained from $y=f(x)$?

a. Discard the portion for $x<0$, and reflect the portion for $x\geq 0$ across the y-axis to form an even function

b. Reflect the entire graph across the x-axis

c. Reflect the graph across $y=x$

d. Shift the graph to the right by $|x|$

32 [Intermediate] • Target: Transformations of Modulus Graphs

What is the vertex (sharp turn) of the function $g(x)=3|x-4|-5$?

a. $(4,-5)$

b. $(-4,-5)$

c. $(4,5)$

d. $(0,-5)$

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33 [Intermediate] • Target: Solving Modulus Equations

Find the oblique asymptote of $f(x) = \frac{x^2 - 14}{(x + 4)}$, in the form $y = mx + c$.

a. $y = 2x - 5$

b. $y = 3x - 5$

c. $y = x - 4$

d. $y = 3x - 2$

34 [Intermediate] • Target: Modulus Inequalities

Find the oblique asymptote of $f(x) = \frac{4x^2 + 3x - 3}{(x)}$, in the form $y = mx + c$.

a. $y = 6x + 4$

b. $y = 4x + 3$

c. $y = 3x + 5$

d. $y = 6x + 1$

35 [Intermediate] • Target: Polynomial Remainder & Factor Theorems

Find the oblique asymptote of $f(x) = \frac{x^2 + 1}{(x + 1)}$, in the form $y = mx + c$.

a. $y = x - 1$

b. $y = 2x + 1$

c. $y = 3x$

d. $y = 2x - 2$

36 [Intermediate] • Target: Transformations of Modulus Graphs

Find the oblique asymptote of $f(x) = \frac{x^2 - 2}{(x - 2)}$, in the form $y = mx + c$.

a. $y = 3x + 3$

b. $y = 2x$

c. $y = x + 2$

d. $y = 2x + 4$

37 [Intermediate] • Target: Solving Modulus Equations

Evaluate $\cos\left(\sin^{-1}(-1)\right)$.

a. -1

b. 0

c. $0 + 3$

d. 2

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Modulus Inequalities • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Evaluate $\cos\left(\sin^{-1}(0)\right)$.

a. 1

b. 0

c. $1+3$

d. -1

39 [Challenging] • Target: Polynomial Remainder & Factor Theorems • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Evaluate $\cos\left(\sin^{-1}(1)\right)$.

a. -1

b. 0

c. 1

d. $0+3$

40 [Challenging] • Target: Transformations of Modulus Graphs • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Solve $|x+1| = |-3x-1|$. Give the largest solution.

a. -2

b. 4

c. 3

d. 0

41 [Challenging] • Target: Solving Modulus Equations • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Solve $|2x+5| = |3x+3|$. Give the largest solution.

a. 6

b. -2

c. 1

d. 2

42 [Challenging] • Target: Modulus Inequalities • Cross-Lesson: 4.02 The Poisson Distribution

Solve $|x+4| = |-2x-2|$. Give the largest solution.

a. 2

b. 5

c. 1

d. 4

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43 [Challenging] • Target: Polynomial Remainder & Factor Theorems • Cross-Lesson: 5.01 Further Integration Techniques

Solve $|x + 3| = |2x + 6|$. Give the largest solution.

a. -1

b. -7

c. -3

d. 0

44 [Challenging] • Target: Transformations of Modulus Graphs • Cross-Lesson: 5.02 Differential Equations

Solve $|x + 3| = |-x + 1|$. Give the largest solution.

a. -1

b. 0

c. -2

d. -5

45 [Challenging] • Target: Solving Modulus Equations • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

Solve $|x - 3| = |-x - 5|$. Give the largest solution.

a. -5

b. 0

c. -2

d. -1

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